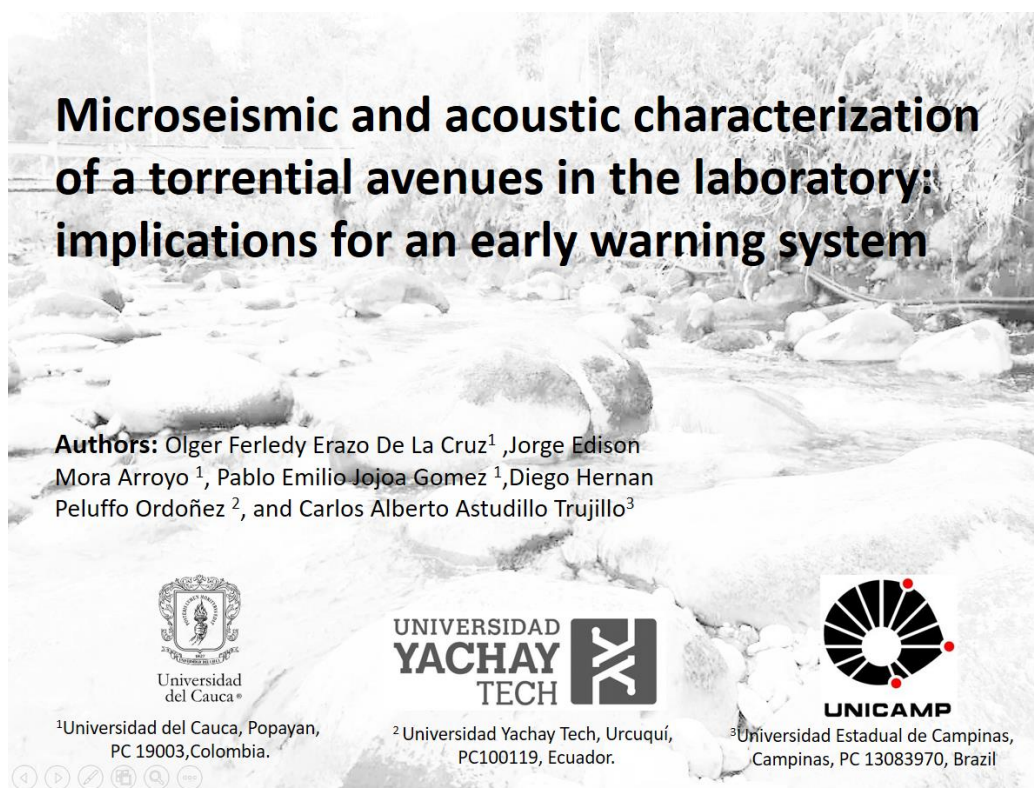


FILE DESCRIPTION



This document contains the description of the files that allow the validation of the results of a classifier of microseismic and acoustic signals generated by a mountain river.

code-python-vbnet-microseismic-acustic-torrential-avenues-laboratory Public Pin Watch 0 Fork 0 Star 0

main 1 Branch 0 Tags Go to file Add file Code

File Name	Action	Time
5_maquina_soporte_vectorial_optuna_sonido_si...	Add files via upload	5 days ago
7_random_forest_optuna_sonido_microsis_16a...	Add files via upload	5 days ago
L_F_04_A_01.rar	Add files via upload	5 days ago
M_F_04_G_01.rar	Add files via upload	5 days ago
README.md	Update README.md	4 days ago
V_F_04_H_01.rar	Add files via upload	5 days ago
calculando métricas desde la matriz de confusi...	Add files via upload	5 days ago
code_pic_c_compiler_data_acquisition_card.rar	Add files via upload	8 minutes ago
code_vb_net_fft_features_level_flow_microseis...	Add files via upload	3 hours ago
features_16agos2026_sismos_audio_unificado.x...	Add files via upload	5 days ago
features_16agos2026_sismos_audio_unificado_...	Add files via upload	5 days ago
install_detects_torrential_avenues_svm.rar	Add files via upload	4 days ago
printed_circuit_components.pdf	Add files via upload	13 minutes ago

About

The repository contains Python code for training SVMs and Random F. using acoustic and microseismic features obtained from experiments involving rock displacement under different flow rates. It includes a CSV file with 78 columns, audio and video files, and experimental data, as well as a module for extracting, merging, and classifying the signals.

Readme
 Activity
 0 stars
 0 watching
 0 forks

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

Microseismic and acoustic characterization of a torrential avenues in the laboratory: implications for an early warning system

Contenido

5_maquina_soporte_vectorial_optuna_sonido_sismos_16ago2026_early_fusion.ipynb	2
7_random_forest_optuna_sonido_microsis_16ago2026.ipynb	2
L_F_04_A_01.rar	2
M_F_04_G_01.rar	3
README.md	3
V_F_04_H_01.rar	4
Calculando_metricas_desde_la_matriz_de_confusion	4
code_pic_c_compiler_data_acquisition_card.rar	5
code_vb_net_fft_features_level_flow_microseisms_acustic_svm.rar	6
features_16agos2026_sismos_audio_unificado.xlsx	7
features_16agos2026_sismos_audio_unificado_csv.csv	10
install_detects_torrential_avenues_svm.rar	11
printed_circuit_components.pdf	13

5_maquina_soporte_vectorial_optuna_sonido_sismos_16ago2026_early_fusion.ipynb

It contains the Python code that loads the dataset from a CSV file, selects sound and microseismic characteristics, chooses different sources, trains the scenario with different parameters using Support Vector Machines; the best hyperparameters are performed with Optuna; the results are displayed in a confusion matrix, there are also functions to export the trained model to C code, which is transpiled to VB .NET.

7_random_forest_optuna_sonido_microsis_16ago2026.ipynb

It contains the Python code that loads the dataset from a CSV file, selects sound and microseismic characteristics, chooses different sources, trains the scenario with different parameters using Random Forest; the best hyperparameters are performed with Optuna; the results are displayed in a confusion matrix, there are also functions to export the trained model to C code, which is transpiled to VB .NET.

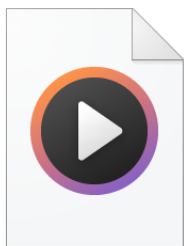
L_F_04_A_01.rar

The folder contains six files related to signals obtained in artificial channels at low flow levels. Under these conditions, there are no tremors or rock impacts. The following image shows the

contents:



L_F_04_A_01_AUDIO_A3A_mic_agua_240104_174433.wav



L_F_04_A_01_AUDIO_A3A_mic_aire_240104_174433.wav



L_F_04_A_01_AUDIO_VIDEO_A3A_240104.wav



L_F_04_A_01_MICROPHONE.csv



L_F_04_A_01_VIBRATION.csv



L_F_04_A_01_VIDEO_A3A_240104.mp4

The above files are used to validate the trained model, which can be run after installing the file on Windows 10 or higher with the filename: install_detects_torrential_avenues_svm.rar

M_F_04_G_01.rar

The folder contains six files related to signals obtained from artificial channels at medium and high flow levels, which sometimes generate tremors. The following image shows the contents:



M_F_04_G_01_AUDIO_mic_agua_A3D1_240105_111846.wav



M_F_04_G_01_AUDIO_mic_aire_A3D1_240105_111846.wav



M_F_04_G_01_AUDIO_VIDEO_A3D1_240105_110855_mono.wav



M_F_04_G_01_MICROPHONE.csv



M_F_04_G_01_VIBRATIONS.csv



M_F_04_G_01_VIDEO_A3D1_240105_110855.mp4

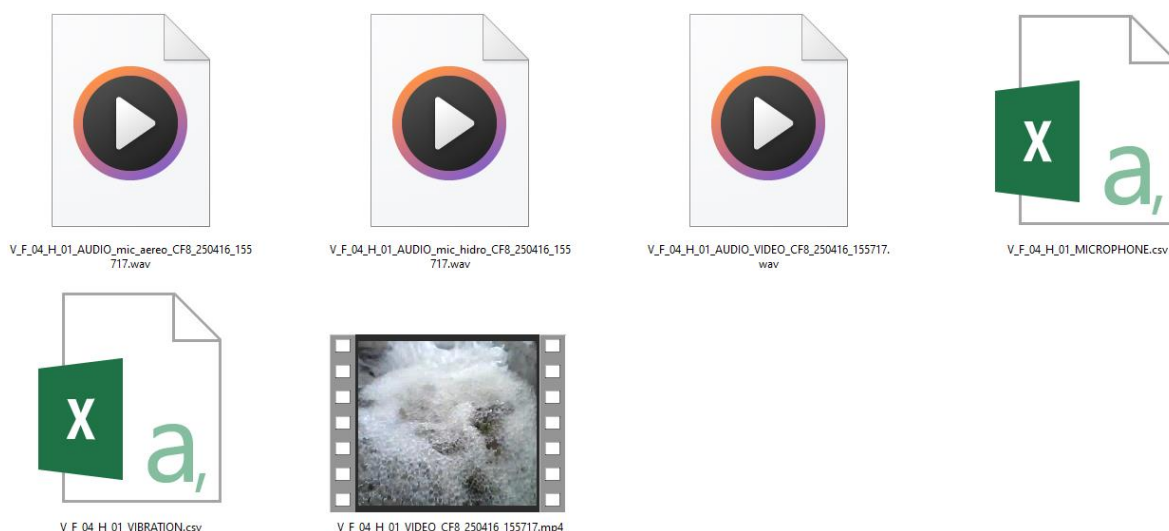
The above files are used to validate the trained model, which can be run after installing the file on Windows 10 or higher with the filename: install_detects_torrential_avenues_svm.rar

README.md

It contains a brief description of the files shared through this medium.

V_F_04_H_01.rar

The folder contains six files related to signals obtained in artificial channels with very high flow rates, which generate tremors and rock impacts. The following image shows the contents:



The above files are used to validate the trained model, which can be run after installing the file on Windows 10 or higher with the filename: install_detects_torrential_avenues_svm.rar

Calculando_metricas_desde_la_matriz_de_confusion

This file contains the metric calculations obtained in the scenarios. Seven scenarios are presented, each with six options. In the latter case, the options relate to the number of features. The file contains seven books. Each book shows the best hyperparameter comparison and the number of features used to train the model.

5 características																
	normal	caudaloso	impacto	precisión no	precisión ca	precisión im	recall no	recall c	recall im	f1-no	f1-ca	f1-im	f1-macro	Accuracy	G-mean	Cohen's kappa
normal	13	1	0	1,000	0,727	1,000	0,929	1,000	0,818	0,963	0,842	0,900	0,902	0,909	0,912	0,863
caudaloso	0	8	0													
impacto	0	2	9													
8 características																
	normal	caudaloso	impacto	precisión no	precisión ca	precisión im	recall no	recall c	recall im	f1-no	f1-ca	f1-im	f1-macro	Accuracy	G-mean	Cohen's kappa
normal	14	0	0	0,933	0,800	1,000	1,000	1,000	0,727	0,966	0,889	0,842	0,899	0,909	0,899	0,861
caudaloso	0	8	0													
impacto	1	2	8													
12 características																
	normal	caudaloso	impacto	precisión no	precisión ca	precisión im	recall no	recall c	recall im	f1-no	f1-ca	f1-im	f1-macro	Accuracy	G-mean	Cohen's kappa
normal	14	0	0	1,000	0,800	1,000	1,000	1,000	0,818	1,000	0,889	0,900	0,930	0,939	0,935	0,908
caudaloso	0	8	0													
impacto	0	2	9													
16 características																
	normal	caudaloso	impacto	precisión no	precisión ca	precisión im	recall no	recall c	recall im	f1-no	f1-ca	f1-im	f1-macro	Accuracy	G-mean	Cohen's kappa
normal	13	1	0	1,000	0,727	1,000	0,929	1,000	0,818	0,963	0,842	0,900	0,902	0,909	0,912	0,863
caudaloso	0	8	0													
impacto	0	2	9													
20 características																
	normal	caudaloso	impacto	precisión no	precisión ca	precisión im	recall no	recall c	recall im	f1-no	f1-ca	f1-im	f1-macro	Accuracy	G-mean	Cohen's kappa
normal	13	1	0	1,000	0,800	1,000	0,929	1,000	0,909	0,963	0,889	0,952	0,935	0,939	0,945	0,908
caudaloso	0	8	0													
impacto	0	1	10													

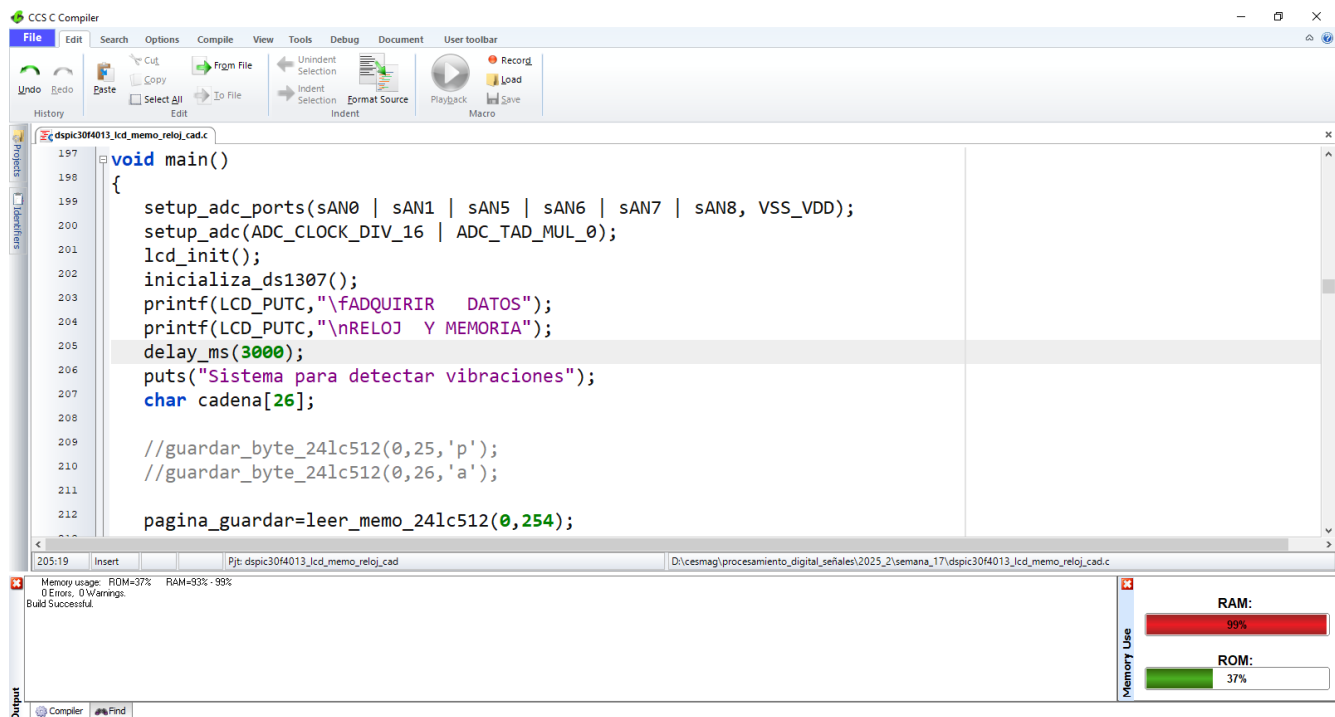
code_pic_c_compiler_data_acquisition_card.rar

This archive contains the source code for the firmware that runs on the acquisition board, which is based on a dsPIC33FJ128MC802 microcontroller.

Inside the folder are source code files and also machine code used to program the microcontroller.

Nombre	Tamaño	Comprimido	Tipo	Modificado	CRC32
..			Carpeta de archivos		
.._history	266.509	69.081	Carpeta de archivos	23/08/2026 5:2...	
dspic30f4013_lcd_memo_reloj_cad.c	17.692	4.367	C Source File	23/08/2026 5:2...	8244293A
dspic30f4013_lcd_memo_reloj_cad.ccsproj	758	339	C Project File	23/08/2026 5:3...	DE6A3523
dspic30f4013_lcd_memo_reloj_cad.cof	252.284	26.412	C Debug File	23/08/2026 5:3...	0910698B
dspic30f4013_lcd_memo_reloj_cad.err	99	99	Archivo ERR	23/08/2026 5:3...	B0C101E5
dspic30f4013_lcd_memo_reloj_cad.esym	54.744	11.937	C Project Output F...	23/08/2026 5:3...	5CE9323A
dspic30f4013_lcd_memo_reloj_cad.h	1.881	1.059	C Include File	20/11/2025 12:...	F58DE096
dspic30f4013_lcd_memo_reloj_cad.hex	66.419	14.713	HEX File	23/08/2026 5:3...	ED408EC1
dspic30f4013_lcd_memo_reloj_cad.lst	183.620	33.350	C/ASM File	23/08/2026 5:3...	8A4F1068
dspic30f4013_lcd_memo_reloj_cad.STA	5.705	1.477	C Stats File	23/08/2026 5:3...	1CE92C4C
dspic30f4013_lcd_memo_reloj_cad.sym	6.781	2.176	C Symbols File	23/08/2026 5:3...	95286041
dspic30f4013_lcd_memo_reloj_cad.tre	16.797	1.195	C Call Tree File	23/08/2026 5:3...	09699167
dspic30f4013_lcd_memo_reloj_cad.xsym	9.770	2.262	C Project Output F...	23/08/2026 5:3...	4104AD09

The code was developed in PIC C Compiler version 5.0. The following figure shows the development environment:



code_vb_net_fft_features_level_flow_microseisms_acustic_svm.rar

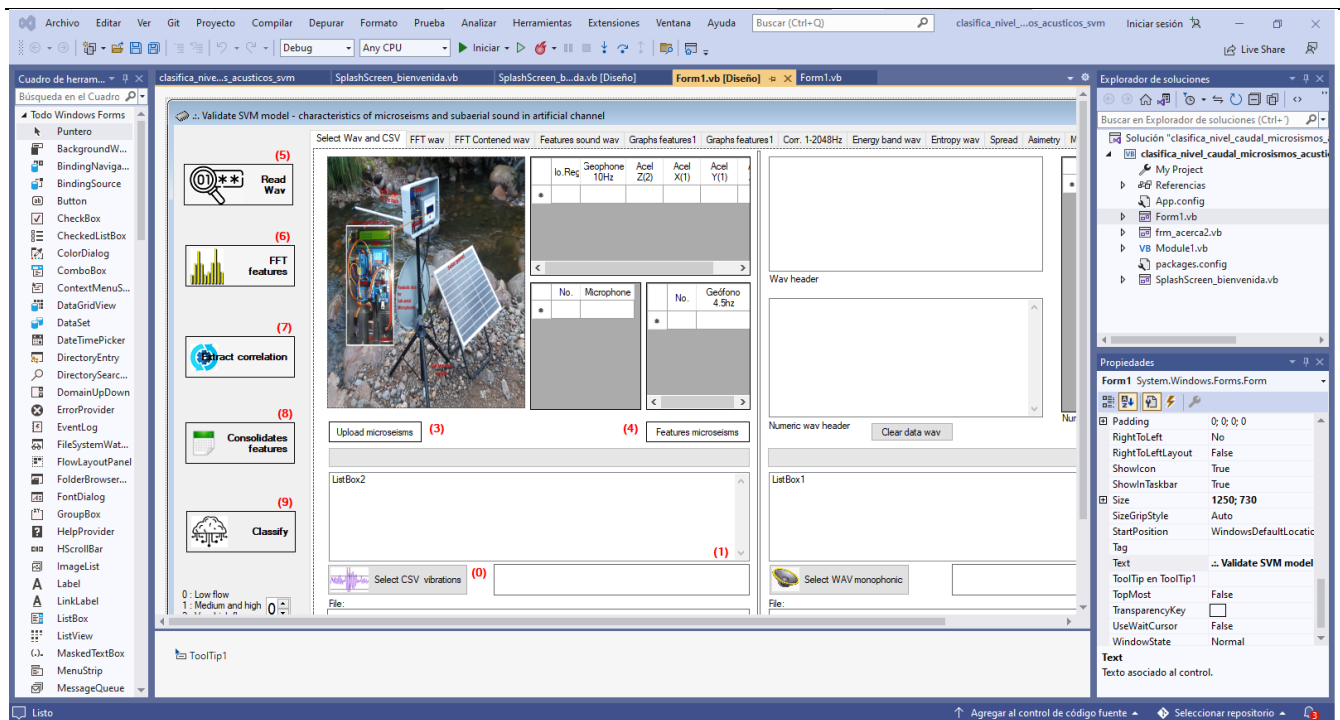
This archive contains code developed in Visual Basic .NET; it allows you to read the contents of WAV audio files and CSV files corresponding to the vibrations captured by the sensors—signals captured by a microphone and a 3-axis analog accelerometer.

Once the files have been selected and read, the FFT is calculated from the matrices, and characteristics in the spectral and temporal domains are estimated, including:

- FFT by bins.
- Container with maximum energy
- Container with minimum energy
- Spectrum correlation.
- Spectral entropy.
- Spectral decay.
- Roll-off.
- Centroid.
- Band-specific energy.
- Energy in the time domain.
- RMS in the time domain.
- Kurtosis.
- Sum of average energy.

Once the features are obtained, each one is normalized using the maximum and minimum values read and recorded in the consolidated features file `features_16agos2026_sismos_audio_unificado.xlsx`.

Once the features are normalized, the model trained using SVM selects the features. Finally, the evaluation results are displayed on the screen.



features_16agos2026_sismos_audio_unificado.xlsx

The file, an XLSX file from Microsoft Office, displays various characteristics, especially descriptive statistical values. It contains 45 characteristics of the spectral domain of the sound generated in the artificial channels, as well as 30 temporal and spectral features of the vibrations. In total, 75 descriptors were obtained from the Visual Basic .NET encoding.

The different characteristics are described below:

a. Features related to energy by band after analyzing 6 observation windows of a 5.4 s audio file, max refers to the maximum value in that band after observing the 6 windows and min indicates the minimum energy recorded in the band after analyzing the 6 observation windows:

C1	C2	C3	C4	C5	C6	C7	C8
max_b1	max_b2	max_b3	max_b4	max_b5	max_b6	max_b7	max_b8

C9	C10	C11	C12	C13	C14	C15	C16
min_b1	min_b2	min_b3	min_b4	min_b5	min_b6	min_b7	min_b8

b. Minimum, average and maximum correlation coefficients. The correlation is estimated after analyzing the spectrum for each of the 6 observation windows. The substring prom refers to average.

C17	C18	C19
coe_cor_min	coe_cor_prom	coe_cor_max

Microseismic and acoustic characterization of a torrential avenues in the laboratory: implications for an early warning system

c. The frequency container with the minimum, maximum, and average energy is estimated. The values obtained are derived from analyzing the 6 windows.

C20	C21	C22
-----	-----	-----

con_min_ene con_max_ene_pro con_max_ene

d. The centroid is estimated for each of the windows, and from the values obtained, the minimum, average, and maximum values are identified.

C23	C24	C25
-----	-----	-----

cen_con_min cen_con_prom cen_con_max

e. The standard deviation between the frequencies of 1 to 1024 Hz and from 1025 to 2048 Hz is also estimated.

C26

des_est_sum_pro

f. The RollOFF at 95% is calculated in each of the 6 windows, then the minimum, average and maximum values are chosen.

C27	C28	C29
-----	-----	-----

rol_inv_min rol_inv_prom rol_inv_max

g. To find out the average energy sum of all containers and in all windows, a value is obtained that is assigned to the C30 characteristic.

C30

sum_ene_pro

h. The spectral entropy is calculated in each of the 6 windows, then the minimum, average and maximum values are chosen

C31	C32	C33
-----	-----	-----

ent_min ent_prom ent_max

i. The 6 features Spread and kurtosis correspond to the estimated values in the 6 observation windows; for each of them, the minimum, average, and maximum values are estimated.

C34	C35	C36	C37	C38	C39
-----	-----	-----	-----	-----	-----

spre_min spre_prom spre_max kur_min kur_prom kur_max

j. The 6 features asymmetry and spectral decay correspond to the estimated values in the 6 observation windows; for each of them, the minimum, average, and maximum values are estimated.

C40	C41	C42	C43	C44	C45
-----	-----	-----	-----	-----	-----

asim_min asim_prom asim_max deca_min deca_prom deca_max

Features of the vibration signal

k. Features related to energy by band after analyzing 6 observation windows of a 5.4 s vibrations file, max refers to the maximum value in that band after observing the 7 windows and min indicates the minimum energy recorded in the band after analyzing the 7 observation windows:

F1	F2	F3	F4	F5	F6
max_b1	max_b2	max_b3	max_b4	max_b5	max_b6

F7	F8	F9	F10	F11	F12
min_b1	min_b2	min_b3	min_b4	min_b5	min_b6

l. The following three time-domain features correspond to the Root Mean Square (RMS), a statistical value that measures the effective magnitude of a signal that changes over time. The seven windows are analyzed, and finally, the minimum, average, and maximum RMS values are estimated from all the observed values.:

F13	F14	F15
rms_min	rms_prom	rms_max

m. The 6 features RollOff at 95% and Centroid correspond to the estimated values in the 7 observation windows; for each of them, the minimum, average, and maximum values are estimated. The substring prom refers to average.

F16	F17	F18	F19	F20	F21
rol_min	rol_prom	rol_max	cen_min	cen_prom	cen_max

n. Minimum, average and maximum correlation coefficients. The correlation is estimated after analyzing the spectrum for each of the 7 observation windows.

F22	F23	F24
corre_min	corre_prom	corre_max

o. The spectral entropy is calculated in each of the 7 windows, then the minimum, average and maximum values are chosen.

F25	F26	F27
ent_min	ent_prom	ent_max

p. The last 3 features correspond to those proposed by the authors to detect vibrations.

F28	F29	F30
-----	-----	-----

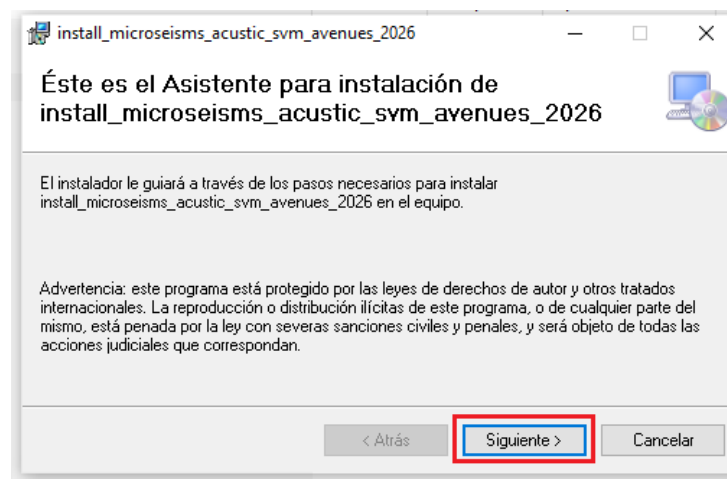
install_detects_torrential_avenues_svm.rar

This archive contains the software installers for validating the classification model. It adds a presentation window and developers to the code included in the file code_vb_net_fft_features_level_flow_microseisms_acustic_svm.rar.

Upon extraction, you will find two files:

install_detects_torrential_avenues_svm.rar\Debug - archivo RAR, tamaño descomprimido 7.305.728 bytes					
Nombre	Tamaño	Comprimido	Tipo	Modificado	CRC32
Carpeta de archivos					
install_microseisms_acustic_svm_avenues_2026.msi	6.730.752	6.346.400	Paquete de Windo...	19/08/2026 11:...	0348E224
setup.exe	574.976	212.070	Aplicación	19/08/2026 11:...	8062EEE6

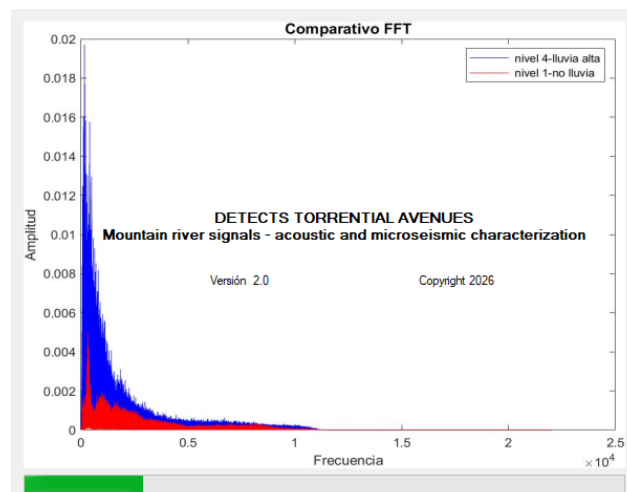
To start the installation, click on the setup.exe file.



Once you press the "Next" button, accept the messages that appear in the following windows.

Finally, run the software and press each button in sequence, from the one labeled 0 to the one labeled 9.

The first screen displayed corresponds to the presentation window:



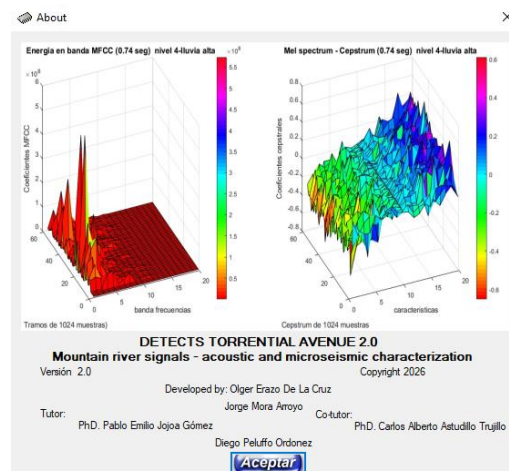
Microseismic and acoustic characterization of a torrential avenues in the laboratory: implications for an early warning system

The main window appears and you click on the button labeled with the number 0; the files to be evaluated are located in the shared folders:

- L_F_04_A_01.rar
- M_F_04_G_01.rar
- V_F_04_H_01.rar

Once you have downloaded the folders containing the audio and vibration signals or accessed the SoMiRiver database, you can evaluate the model by running the following form.

The developer window appears when you click the button labeled “About us”, bottom left corner of the form. The system displays the following window:



Microseismic and acoustic characterization of a torrential avenues in the laboratory: implications for an early warning system

